



Sharayu Mhaske

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About Me

Passionate **Angular Developer** and **AI/ML enthusiast** with professional experience at **Fraunhofer FKIE, Germany** and expertise in building scalable **B2C web applications**. Specializes in **Angular, RxJS, Firebase, and REST API integration**. Actively works in **Artificial Intelligence** — fine-tuning **NLP models** like **RoBERTa** and **DistilBERT** and conducting published research in **Responsible AI** at **ICAAI '25**. Currently completing a **Master's in Web and Data Science** at **Universität Koblenz**, with a thesis on **Emotion Detection in Social Media**. Seeking a **full-time role** to contribute across the full stack within a dynamic, innovation-driven team.

Experience

Student Assistant

May 2024 – Present

Fraunhofer FKIE

Bonn, Germany

- Architected a **dynamic tree component** using **Angular Material** (Mat-Tree, MatDialog, MatFormField) to render and edit nested **JSON data structures**.
- Implemented **Angular CDK Drag & Drop** for real-time hierarchical node reordering, supporting parent-child restructuring and **immutable state updates**.
- Built scalable **Reactive Forms** using **FormBuilder, FormGroup, and Validators** for dynamic node creation, inline editing, and **form validation**.
- Enabled **cross-component communication** via **Angular EventEmitter** and **@Input/@Output decorators** to propagate state changes across a reactive **SPA**.
- Managed **application state** using **RxJS BehaviorSubject** and **ReplaySubject** within **Angular Services**, enabling real-time **UI synchronization**.
- Integrated **RESTful APIs** using **Angular HttpClient** with typed **HTTP requests** to serialize and persist **JSON payloads** to backend services.
- Versioned and collaborated on all features using **Git** and **GitHub**, managing **feature branches, pull requests, and code reviews**.
- Expanding **full-stack capabilities** by building **Java Spring Boot** backend services and consuming **REST API contracts** integrated with the **Angular frontend**.

Software Development Intern

Sep 2021 – Mar 2022

Star Spring Works Pvt. Ltd.

Mumbai, India

- Developed and maintained multiple **Angular** applications for **B2C** clients across various domains.
- Implemented **FACADE architecture** to abstract and centralize **API service calls**, improving code maintainability.
- Integrated the **Camera Module** to capture and upload images, storing **Blob data** in **Firestore** using **AngularFire**.
- Built a lightweight **custom state management store** using **RxJS** and **BehaviorSubject** for shared application state.
- Wrote and maintained **Unit Test Cases** using **Jasmine/Karma** to support a stable **CI/CD pipeline**.
- Managed source code using **Git** and **GitHub**, handling **version control** and **feature branching**.
- Responsible for **building and deploying** Angular applications to client domains using **Angular CLI**.

Education

Master's in Web and Data Science

Oct 2023 – Present

University of Koblenz

Koblenz, Germany

Bachelor's in Computer Engineering

Jun 2017 – May 2021

University of Mumbai

Mumbai, India

Projects

Master Thesis — Emotion Detection in Social Media

May 2026 – Present

- Extended **SemEval Task 11 (2025)** using the **BRIGHTER dataset** to compare single-step vs. two-step emotion detection and intensity pipeline approaches.
- Fine-tuned **RoBERTa** and **DistilBERT** to determine if emotion and intensity can be predicted jointly or require separate stages.
- Trained models across multiple batch sizes using **k-fold cross-validation** and **scikit-learn**.
- *Tech Stack: RoBERTa, DistilBERT, Hugging Face Transformers, scikit-learn, Python, Pandas, NumPy.*

Bias in Recruitment Systems Utilizing Large Language Models

Oct 2024 – Feb 2025

- Analyzed **gender, racial, and geographical biases** in an **AI-driven recruitment system** using a Recruitment Dataset.
- Adapted the **Word Embedding Association Test (WEAT)** to quantify **selection bias** across demographic groups in recruitment **AI/ML models**.
- Generated **WEAT score visualizations** exposing systematic hiring preferences across candidate demographics.
- Managed research codebase using **Git** and **GitHub**; co-authored and published findings in **ICAAI '25** proceedings, contributing to **Responsible AI** and **Algorithmic Fairness** research.

Text Summarization Using Deep Learning

- Built a **text summarization pipeline** preprocessing large **NLP datasets** using **Pandas** and **NumPy**, handling null values, sequence padding, and start/end tokens.
- Trained and evaluated a **Seq2Seq** summarization model using **word embeddings** and **recurrent architectures** to generate concise summaries from long-form text.
- Managed project codebase using **Git** and **GitHub** across the full **ML pipeline**.

Kanban Board for Project Management

- Built a **Kanban-based task management** app using **Django REST Framework** for backend **REST APIs** and **Angular** for the frontend.
- Implemented secure **login, signup, and token-based authentication** APIs using **Django's authentication** system.
- Developed full **CRUD operations** for task management across Kanban board columns using **REST API endpoints**.

Rock, Paper, Scissors — Web Game

- Built an interactive **web-based game** using **HTML, CSS, JavaScript, and Angular** with real-time user input and randomized computer responses.
- Implemented **game logic** and **event-driven interactions** managing win/loss/draw conditions using **Angular components**.

Technical Skills

- **Languages:** Python, JavaScript, TypeScript, Java, C, SQL, PHP
- **Frontend:** Angular 17+, RxJS, Angular Material, Angular CDK, HTML5, CSS3, React.js
- **Backend:** Node.js, Express.js, FastAPI, Django REST Framework, Java Spring Boot
- **AI/ML:** RoBERTa, DistilBERT, BERT, GPT, Hugging Face Transformers, Scikit-learn, TensorFlow, Pandas, NumPy
- **Databases:** MySQL, Firebase Firestore, MongoDB, SQL
- **DevOps & Tools:** Git, GitHub, GitLab, Angular CLI, Jasmine/Karma, CI/CD Pipelines
- **Concepts:** REST APIs, Reactive Programming, State Management, NLP, Responsible AI, Algorithmic Fairness

Languages

Marathi: Native **English:** C2 – Proficient **Hindi:** C2 – Proficient **German:** B1 – Independent

Certifications

Complete Web Development Bootcamp – Udemy
Introduction to Python Programming – Microsoft
Complete Guide to TensorFlow with Python – Udemy
Intro to Machine Learning – Kaggle

Achievements

- **Published Research Paper** - *Bias in Recruitment Systems Utilizing Large Language Models*, ICAAI 2025: 9th International Conference on Advances in Artificial Intelligence.
- **First Prize** in Technical Project Presentation at **XTACY 2k18** - Project: Smart Glasses.
- **GATE Qualified** - 2022.